Analytics Trainee Assessment

FinSight Lanka - Customer Data Analysis

Confidential - For Assessment Purposes Only

Background

FinSight Lanka is a fast-growing fintech company offering digital savings accounts, personal loans, and a mobile wallet to retail customers across Sri Lanka. Over the past two years, the company has expanded rapidly across all provinces, acquiring customers through digital channels, branch networks, agents, and referrals.

As the company scales, the leadership team needs to make informed decisions on customer risk, product cross-selling, digital adoption, and regulatory compliance. You have been brought in as an Analytics Trainee and given a dataset of 500 customer records. Your job is to clean, analyse, and present meaningful insights that can inform real business decisions.

Important Note on Data Quality

> The data set provided has been prepared to reflect real-world conditions. It intentionally contains a range of data quality issues, including but not limited to duplicate records, missing values, impossible entries, formatting inconsistencies, and contradictions across columns. This is not accidental.
>
> In the real world, raw data is almost never clean. A large part of an analyst's job is identifying and handling these issues before drawing any conclusions. You are expected to identify, document, and address these issues as part of your analysis. Findings drawn from uncleaned data will be considered incomplete.
>
> **There is no single correct answer - we are evaluating your thought process, your judgment in handling ambiguous data, and your ability to turn raw numbers into clear business insights.**

Assessment Questions

Q1 - Savings & Customer Behaviour

- Analyse average savings balance across Customer Segment, Province, and Urban/Rural classification. Highlight any meaningful differences between groups.
- Which age group holds the highest total savings? Define your own age bands, explain your banding logic, and support your answer with data.
- What percentage of customers are actively growing their savings versus draining them each month? What does this suggest about the health of the customer base?

Q2 - Loan Portfolio & Risk

- What is the overall loan penetration rate? Break it down by Segment and Acquisition Channel and identify where the highest concentration lies.
- Calculate the Debt-to-Savings Ratio for all customers with loans (Outstanding Loan Balance ÷ Savings Balance). How many customers exceed a ratio of 1.0 and what is the total LKR exposure this represents?
- Is there a pattern in loan default rates across Segment, District, or Loan Type? Flag any combinations that represent elevated risk.

Q3 - Digital Engagement & Product Holdings

- Do mobile app users maintain higher savings balances on average? Support your answer with a clear comparison and state whether you believe the relationship is meaningful.
- Identify customers who hold savings but have no Fixed Deposit and no Insurance. Quantify this cross-sell opportunity in both customer count and total savings value held by this group.
- Which Acquisition Channel consistently produces customers with the highest savings balance and the lowest default rate? What does this imply for business?

Q4 - Define 5 KPIs

Define 5 KPIs that FinSight Lanka should track monthly to measure the health of its customer base. For each KPI provide:

- A clear name and one-line definition
- The formula or calculation using fields from the dataset
- The current value calculated from your cleaned data
- Why it matters to the business and what decision it informs
- A recommended target or benchmark with justification

Your 5 KPIs must span at least 3 different areas such as risk, growth, engagement, compliance, or retention. *(Generic KPIs without a formula or business rationale will not score well)*

Dashboard note: These KPIs should be presented visually in a dashboard format - using Tableau, Power BI, or a similar tool. The dashboard should be designed to communicate your KPIs clearly to a non-technical business audience. A PDF export or screenshot of your dashboard must be included as part of your submission.

Q5 - Board Summary & Recommendation

- Summarise your 3 to 4 most important findings in plain business language - no technical jargon.
- Make two specific strategic recommendations clearly backed by your data.
- State two additional data sources you would want access to and explain why they would strengthen the analysis.

Q6 - Critique Your Own Work

Step back and review your own analysis critically. Produce at least 3 specific criticisms of your own conclusions, these could relate to data quality assumptions you made, analytical choices that could be questioned, variables that are missing, or findings that could be interpreted differently.

- For each criticism, state clearly what the limitation is and rate its severity as High, Medium, or Low.
- Explain why it matters - how does it affect the reliability or usefulness of your findings?
- Suggest how you would address it if you had more time, more data, or a different approach.

Strong candidates go beyond the obvious. Identifying that "the dataset is small" is not enough, we want to see you think carefully about the specific weaknesses in your own reasoning and conclusions.

Deliverables & Submission

Item	Requirement
Workings	Excel workbook, Python / Jupyter Notebook, or Tableau / Power BI file with all analysis and data cleaning steps clearly labelled by question number.
Presentation	PowerPoint slide deck - maximum 10 slides Slides should tell the story of your analysis and be suitable for a non-technical business audience.
Dashboard	A PDF export or screenshot of your KPI dashboard built in Tableau, Power BI, or a similar tool.
**If you are unable to attach all the above documents to the email, please upload them to Microsoft OneDrive or Google Drive and share the link via email. Kindly ensure that the folder is publicly accessible through the shared link.*	
File Naming	Workings: FirstName_LastName_FinSightAssessment_Workings Presentation: FirstName_LastName_FinSightAssessment_Presentation Dashboard: FirstName_LastName_FinSightAssessment_Dashboard
Submission	**Email all files to Pumudika.Amarasekara@dialogfinance.lk** **Subject: Analytics Trainee Assessment - [Your Name]**